

Illuminating the Interactome with BEACON

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Abstract

Protein-protein interactions and functional associations are a powerful way to identify proteins working in the same pathways or sharing similar biological functions, which when disrupted are more likely to lead to similar phenotypes and disorders. While these interactions are typically viewed qualitatively, using network viewers, a quantitative approach to these functional associations would provide a way to measure the closeness of interactions between a set of proteins and rank candidate gene lists by their functional similarity to known disease-associated proteins. Toward this, we have developed BEACON, a method using degrees of separation of interactions between proteins to generate quantitative scores of network interconnectivity. Using BEACON built on functional associations from the STRING database we found consistent performance of BEACON numbers across varying confidence thresholds and demonstrated its application to human phenotypes by prioritizing candidate genes for a severe subphenotype of autism.

Intro

The study of rare complex genetic diseases represents a key frontier in advancing human health for cases which are disproportionately debilitating, pediatric, and lethal (Qi *et al.*, 2024; Chung *et al.*, 2022). In complex disease the implementation of next generation sequencing technologies, including whole genome and whole exome sequencing, have greatly improved the identification of disease candidate genes, however validating the association between candidate gene and disease remains a significant challenge (Qi *et al.*, 2024). One method to identify potential disease candidate genes involves examining genes containing *de novo* variants, particularly loss-of-function variants (frameshift, nonsense, and splice site variants) which are more likely to result in reduced or ablated protein function for a phenotype of interest (MacArthur and Tyler-Smith, 2010). These candidate genes lists may reveal individual genes of interest using curated methods functional annotations such as Gene Ontology and protein protein interaction (PPI) networks, but are also likely to contain numerous false positives and fail to highlight genes without known function relating to the studied trait (Hart *et al.*, 2006; Wu *et al.*, 2009). Developing additional tools to further investigate and refine candidate gene lists will be instrumental to advancing our understanding of rare diseases.

Here, we introduce BEACON, a method to quantify the interconnectivity of a candidate gene network by measuring the shortest path of functional associations between proteins from the STRING database (Szkarczyk *et al.*, 2025).

BEACON

Studies of degrees of separation are traditionally found in studies of social interaction, however this framework may also benefit the study of systems biology scope of the protein interactome (Rooke *et al.*, 2024; Vidal *et al.*, 2011; Huttlin *et al.*, 2015). Genes associated with the same complex disease are significantly more likely to be in the same PPI network than with genes associated with other diseases (Barrenas *et al.*, 2009; Goh *et al.*, 2007). This enrichment is partially explained by the idea of cascading effects, where mutations in distinct proteins within a network can lead to the same disrupted downstream processes (Guimarães *et al.*, 2018; Posey *et al.*, 2017). Indeed, many human diseases sharing phenotypic overlap arise from mutations (Rosenthal *et al.*, 2023) within the same pathway including the neuro-cardio-facial-cutaneous syndromes of the RAS pathway (Bentires-Alj *et al.*, 2006), cancers of the receptor tyrosine kinase pathways (Vogelstein and Kinzler, 2004), and the cell cycle pathway in Alzheimer's (Yurov *et al.*, 2011). By integrating PPI network analysis into the study of complex disease we can analyze the interconnectedness of candidate genes to known disease-associated genes providing additional evidence of their role in a disease pathway.

Previous methods have utilized PPI network data to further refine our understanding of the interactome or identify candidate genes. Many tools to refine candidate lists have been developed, including tools primarily using text mining such as MinMiner (van Driel *et al.*, 2006) and GeneProspector (Yu *et al.*, 2008), generating subnetworks such as LEAN (Yu *et al.*, 2008; Gwinner *et al.*, 2017; Ideker *et al.*, 2002), and those that identify novel candidate genes through the assessment of known disease genes such as CPS (George *et al.*, 2006), phenoPhred (Radivojac *et al.*, 2008), and NetColoc which identifies candidate genes co-localized from two diseases (Rosenthal *et al.*, 2023). Newer methods of PPI refinement include additional data types such as gene expression data (Shah and Braun, 2019) or GWAS data (Greene *et al.*, 2015), which may not be available in the case of rare disorders. Methods that do measure the interconnectedness of a network typically only look at direct neighbors, potentially missing proteins in the same pathway further up-or-downstream (Oti *et al.*, 2006; Gwinner *et al.*, 2017; Ostlund *et al.*, 2010). Together, these methods all provide evidence for the importance of PPI networks in identifying additional genes contributing to a similar phenotype, however they rarely consider the interactome in its entirety.

In order to address this gap, we have developed BEACON, a tool designed to quantify the association strength between proteins by identifying the shortest number of functional links between them in a STRING PPI network. We chose the STRING database for its incorporation of a wide range of both physical and functional interaction evidence, including data from multiple PPI sources, gene ontology, and coexpression (Szklarczyk *et al.*, 2025). The current iteration of STRING contains over 50 million unique proteins with interactions curated from literature searches, other databases, and predicted using computation methods. By utilizing all lines of STRING evidence, BEACON identifies the shortest path of functional association edges between all reported protein-protein pairs using breadth first path finding. When provided with a protein list, an overall BEACON number is determined by averaging the number of links between all proteins in the list, quantifying the overall interconnectedness of the given proteins (Figure 1A). By

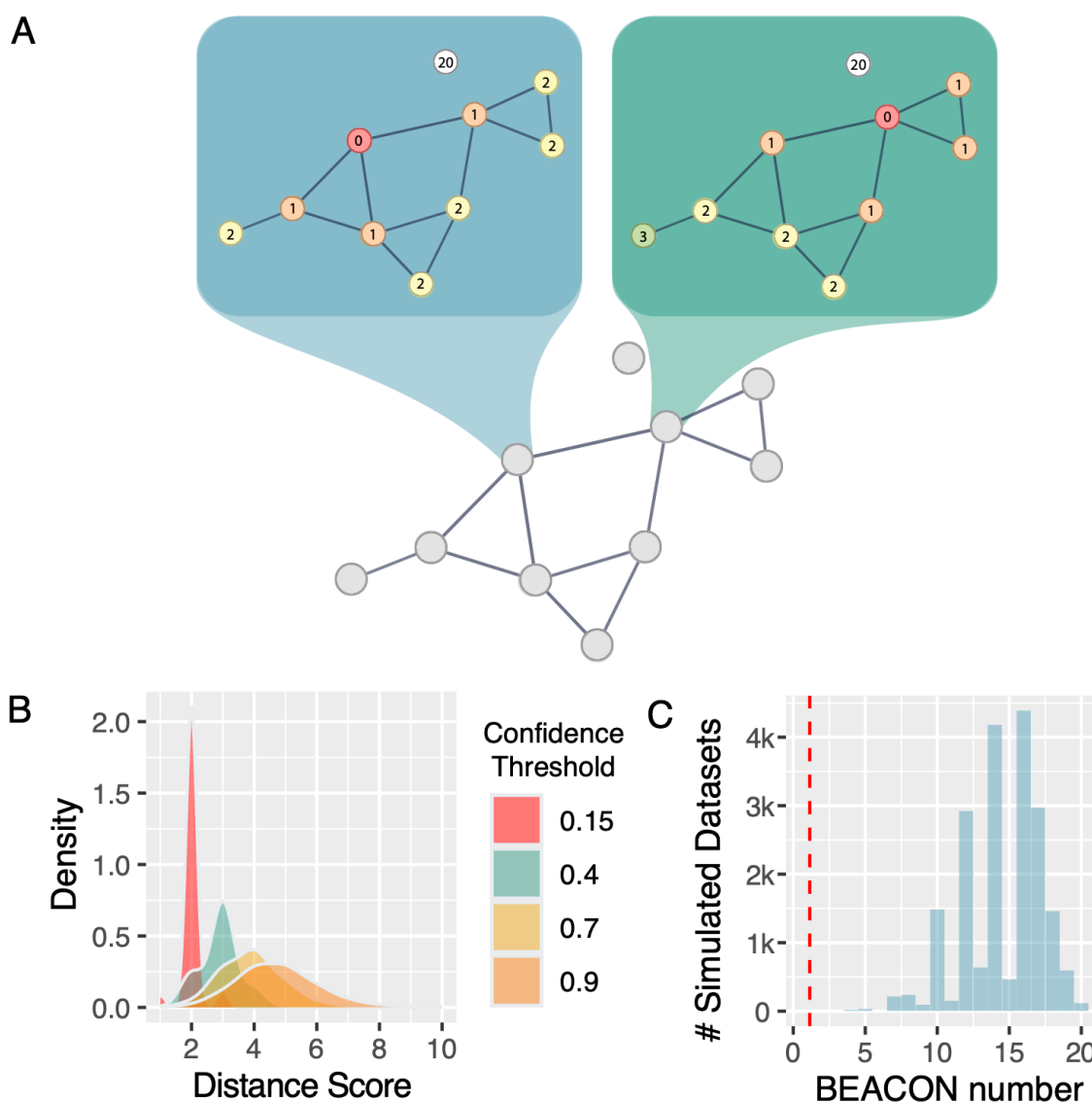


Figure 1. A. Example STRING network showing proteins of differing degree distances within the same network, where the number on the node represents the distance from the calculated protein (red). B. Distributions of distance scores for all 4 confidence thresholds. C. Positive control score plotted against random genes (dashed red line) using the 0.4 threshold (monte cristo approximation example).

processing individual proteins alongside known disease genes, candidate genes can be ranked by BEACON number, assisting with prioritization of candidates for experimental validation.

BEACON works by viewing the interactome as a whole to examine the closeness of all pairwise protein-protein nodes via path finding. Cached pairwise distance values correspond to the number of edge links between the two proteins based on the functional data provided. A value slightly larger than the largest number of steps (20) was chosen as the punitive score given to protein pairs with no known interaction path. All pairwise values can then be averaged across a candidate list to determine the list's BEACON number, theoretically ranging from 1-20, where 1 would

represent a network where every protein interacted with every other protein in a list, and 20 would be a network with no known interactions between any protein pairs.

BEACON has built-in options to facilitate user chosen confidence thresholds, which indicate the expected percentage of false positive interactions in a network (Szkłarczyk *et al.*, 2025). Though any value between 0-1 is accepted by BEACON, in this paper we explored the 4 thresholds defined by STRING: 0.9 highest, 0.7 high, 0.4 medium, 0.15 low. In order to determine if a given gene list is more interconnected than expected, lists of randomly chosen proteins from the STRING database of equal length to the candidate list are generated. BEACON numbers were then calculated for 20,000 of these simulated datasets followed by a Monte Carlo approximation to obtain a p-value. Output files include a list of all BEACON numbers for the random protein lists used for statistical analysis and all pairwise interaction numbers corresponding to the shortest number of edges between each protein pair for the candidate protein list.

In order to assess how BEACON performs using different parameters we first ran our pathway finding algorithm to generate pairwise distance values for networks of different STRING confidence thresholds. As expected, the number of direct interactions using the full STRING network revealed greater numbers of direct interactions at lower confidence thresholds (0.9=67mil, 0.7=126mil, 0.4=180mil, 0.15=192mil). Comparing the topology of these networks, the distribution of distance scores of each threshold appears much more dispersed for higher confidence thresholds (Fig 1B), and the number of disconnected subnetworks shows a striking decrease going from high to low confidence thresholds, with all proteins in the low confidence threshold contained within a single network (0.9=7.5k, 0.7= 3.5k, 0.4=136, 0.15=1) (Sup Fig 1). We found little change between the mean of BEACON numbers of simulated networks of different lengths ranging from 10-400 proteins using a medium confidence score, however standard deviations fell considerably up to lists of 400 proteins, suggesting more consistent levels of significance for larger protein inputs (Sup 2).

SMO and the hedgehog pathway

We examined the example STRING network for the protein 'smoothened' (SMO) as a positive control for a known gene network using the options of a full STRING network, medium confidence (0.4), and no more than 10 interactors (ARRB1, ARRB2, GLI1, GRK2, IHH, KIF3A, PTCH1, PTCH2, SHH, SUFU). SMO is a key regulator of the hedgehog pathway, activation of which is known to contribute to many types of cancer (Lou *et al.*, 2020). We extracted ENSEMBL protein IDs from the node2_string_id column of the exported tsv file from STRING and ran BEACON at the 4 STRING confidence thresholds. All BEACON numbers recovered (0.9=1.45, 0.7=1.32, 0.4=1.14, 0.15=1.01) were lower than all generated random BEACON numbers, yielding high levels of significance (0.9=14.59, 0.7=9.47, 0.4=3.14, 0.15=2.01; Monte Carlo approximate p-values 0.9=4.9*10⁻⁵, 0.7=4.9*10⁻⁵, 0.4=4.9*10⁻⁵, 0.15=4.9*10⁻⁵) (Fig 1C).

Known Functional Human Gene Networks from OMIM

We next examined the BEACON numbers of annotated lists of genes associated with human disease from the Online Mendelian Inheritance in Man (OMIM) as curated by Franke *et al.*, 2006 (Sup Table 1) (Amberger *et al.*, 2015; Franke *et al.*, 2006). Out of the 110 diseases all but three (Bile-acid synthesis defect congenital, Cutis laxa, and Microphthalmia) had significantly lower BEACON scores compared to a random gene list of the same size (Monte Carlo approximate p-values 0.07, 1.4×10^{-4} , and 1.4×10^{-4} respectively). These cases are all driven by individual proteins outside the larger network. In the case of congenital Bile-acid synthesis defect, impaired bile synthesis leading to poor fat absorption, the reduced score is driven by AMAC, associated with autosomal recessive type 4 bile acid synthesis (Wang *et al.*, 2021). The syndromic subtype of Microphthalmia, a structural eye disorder, has three proteins with no connectivity within the network (BCOR, HCCS, and SNX3) (Plaisancie *et al.*, 2016). Both BCOR and HCCS are associated with syndromic versions of Microphthalmia, and SNX3 has since been shown to be a spurious finding (Kumar *et al.*, 2007). Interestingly, for Cutis laxa the protein outside of the network (ATP7A) represents the entirety of the X-linked recessive subphenotype. This rare subphenotype shows a distinct phenotypic presentation including generalized skin laxity and a distinct facial appearance (Gardeitchik *et al.*, 2014). For all 3 OMIM diseases showing unexpectedly high BEACON scores, our findings suggest that further validation and research into differing mechanisms of disease would be of interest in these cases.

Additionally, we compared the BEACON numbers for seven genes associated with familial breast cancer from the OMIM database to known lifetime risk breast cancer percentages by exclusion analysis (Figure 2) (Roberts *et al.*, 2023). Least square regression analysis shows high agreement with coefficient of determination (R^2) of 0.86. This suggests BEACON scores are able to correctly rank risk genes in the presence of a known PPI network.

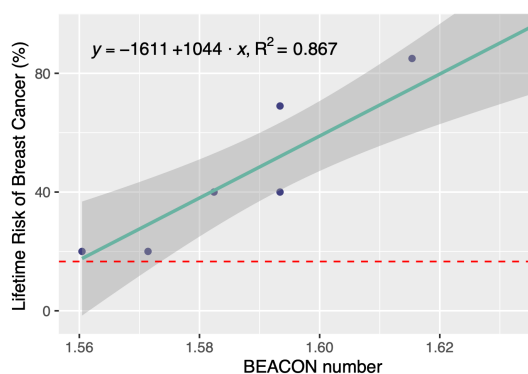


Figure 2. Linear regression analysis of familial breast cancer BEACON scores from exclusion analysis compared to percentage known lifetime breast cancer risk (Roberts, 2023). An R^2 of 0.86 shows a high agreement between the 2 scores. The red dashed line represents the background lifetime breast cancer risk.

Candidate genes in a severe sub-phenotype of autism

Autism presenting with disproportionate megalencephaly (ASD-DM) - an enlarged brain, represents an enriched severe sub-phenotype of autism (Amaral *et al.*, 2017). While 15% of ASD-DM cases may be explained by mutations in two genes (*CHD8* & *PTEN*) further research is required to unravel the full genetic etiology of ASD-DM [Wu, 2020]. A recent study identified 154 ASD-DM candidate genes containing *de novo* loss-of-function variants from trio next generation sequencing data (Nishizaki *et al.*, 2025). Though this study integrated network analysis and gene ontology information into their study, the lack of ranked evidence from these analyses led them to choose genes for experimental follow-up based on literature review and assessment method.

By running the entire ASD-DM candidate gene list through BEACON we found the resulting proteins form a significantly more interconnected network than would be expected by chance at every STRING confidence threshold (0.9=11.87, 0.7=8.04, 0.4=2.72, 0.15=1.01; Monte Carlo approximate p-values 0.9=3.4*10⁻⁴, 0.7=0.04, 0.4=1.4*10⁻⁴, 0.15=4.9*10⁻⁵).

Following this, we included an additional 16 genes harboring *de novo* variants related to larger head circumference in autism as a known association network and integrated the candidate genes one-by-one to generate ranked BEACON numbers for each of the 153 candidate ASD-DM genes (Wu *et al.*, 2020). Of note, the top 7 hits not overlapping the known associate network have been previously associated with either autism or large head phenotypes, suggesting genes associated with similar phenotypes are more likely to have a lower BEACON number, and thus a high ranking in our analysis (Sup Table 2).

Availability and Implementation

BEACON is implemented in Python 3.8.19 and Perl 5.34.1 and distributed under a GPL-3.0 License. BEACON is available on github at: <https://github.com/sierranishizaki/BEACON>.

Conclusion

PPI analysis methods are useful in determining potential shared mechanisms and pathways of candidate proteins. However, available methods to quantify the interactome as a whole remain elusive. BEACON uses breadth first path finding to quantify the interconnectivity of an entire network, expanding our understanding of distinct mechanisms in complex diseases. Using BEACON we can also examine changes in interconnectivity of a network when individual genes are introduced, providing a powerful tool for parsing candidate genes lists in rare diseases.

Though useful, there remain challenges to utilizing PPIs as a basis for further advances. Because the majority of PPI networks are built on curated evidence of protein interactions, leaving them biased toward well annotated genes, and may be incomplete and error-prone (Wu *et al.*, 2009). While STRING partially alleviates this by integrating PPI data from multiple sources and orthologous data from other species, this bias is unavoidable (Szklarczyk *et al.*, 2025).

Despite this bias it has been predicted that 10% of proteins interacting with a disease gene are likely to contribute to the same disease (Oti *et al.*, 2006). There have been developments to impute protein interactions using machine learning, greatly expanding our predictions of protein mechanisms, structure, and biological pathways (Muzio *et al.*, 2021). These advances will require experimental validation leaving a greater need for tools that can assist with candidate gene ranking, and may not be available in cases of rare disease with few known contributing genes.

In a dearth of methodologies appropriate for rare disease, BEACON bridges the gap between candidate gene discovery and gene function by examining the connectivity of candidate genes.

References

- Amaral,D.G. *et al.* (2017) In pursuit of neurophenotypes: The consequences of having autism and a big brain. *Autism Res*, **10**, 711–722.
- Amberger,J.S. *et al.* (2015) OMIM.org: Online Mendelian Inheritance in Man (OMIM®), an online catalog of human genes and genetic disorders. *Nucleic Acids Res*, **43**, D789–98.
- Barrenas,F. *et al.* (2009) Network properties of complex human disease genes identified through genome-wide association studies. *PLoS One*, **4**, e8090.
- Bentires-Alj,M. *et al.* (2006) Stops along the RAS pathway in human genetic disease. *Nat Med*, **12**, 283–285.
- Chung,C.C.Y. *et al.* (2022) Rare disease emerging as a global public health priority. *Front Public Health*, **10**, 1028545.
- van Driel,M.A. *et al.* (2006) A text-mining analysis of the human phenome. *Eur J Hum Genet*, **14**, 535–542.
- Franke,L. *et al.* (2006) Reconstruction of a functional human gene network, with an application for prioritizing positional candidate genes. *Am J Hum Genet*, **78**, 1011–1025.
- Gardeitchik,T. *et al.* (2014) Clinical and biochemical features guiding the diagnostics in neurometabolic cutis laxa. *Eur J Hum Genet*, **22**, 888–895.
- George,R.A. *et al.* (2006) Analysis of protein sequence and interaction data for candidate disease gene prediction. *Nucleic Acids Res*, **34**, e130.
- Goh,K.-I. *et al.* (2007) The human disease network. *Proc Natl Acad Sci U S A*, **104**, 8685–8690.
- Greene,C.S. *et al.* (2015) Understanding multicellular function and disease with human tissue-specific networks. *Nat Genet*, **47**, 569–576.
- Guimarães,P.R.,Jr *et al.* (2018) Interaction paths promote module integration and network-level robustness of spliceosome to cascading effects. *Sci Rep*, **8**, 17441.
- Gwinner,F. *et al.* (2017) Network-based analysis of omics data: the LEAN method. *Bioinformatics*, **33**, 701–709.
- Hart,G.T. *et al.* (2006) How complete are current yeast and human protein-interaction networks? *Genome Biol*, **7**, 120.
- Huttlin,E.L. *et al.* (2015) The BioPlex Network: A Systematic Exploration of the Human Interactome. *Cell*, **162**, 425–440.
- Ideker,T. *et al.* (2002) Discovering regulatory and signalling circuits in molecular interaction networks. *Bioinformatics*, **18 Suppl 1**, S233–40.
- Kumar,R.A. *et al.* (2007) Absence of mutations in NR2E1 and SNX3 in five patients with MMEP (microcephaly, microphthalmia, ectrodactyly, and prognathism) and related phenotypes. *BMC Med Genet*, **8**, 48.
- Lou,H. *et al.* (2020) Genetic and Epigenetic Regulation of the Smoothed Gene (SMO) in Cancer Cells. *Cancers (Basel)*, **12**.
- MacArthur,D.G. and Tyler-Smith,C. (2010) Loss-of-function variants in the genomes of healthy humans. *Hum Mol Genet*, **19**, R125–30.
- Muzio,G. *et al.* (2021) Biological network analysis with deep learning. *Brief Bioinform*, **22**, 1515–1530.
- Nishizaki,S.S. *et al.* (2025) mA-mRNA Reader YTHDF2 Identified as a Potential Risk Gene in Autism With Disproportionate Megalencephaly. *Autism Res*, **18**, 966–982.
- Ostlund,G. *et al.* (2010) Network-based Identification of novel cancer genes. *Mol Cell Proteomics*, **9**, 648–655.
- Oti,M. *et al.* (2006) Predicting disease genes using protein-protein interactions. *J Med Genet*, **43**, 691–698.
- Plaisancie,J. *et al.* (2016) Genetic Advances in Microphthalmia. *J Pediatr Genet*, **5**, 184–188.
- Posey,J.E. *et al.* (2017) Resolution of Disease Phenotypes Resulting from Multilocus Genomic Variation. *N Engl J Med*, **376**, 21–31.
- Qi,T. *et al.* (2024) From genetic associations to genes: methods, applications, and challenges. *Trends Genet*, **40**, 642–667.
- Radivojac,P. *et al.* (2008) An integrated approach to inferring gene-disease associations in humans. *Proteins*, **72**, 1030–1037.
- Roberts,E. *et al.* (2023) Polygenic risk scores and breast cancer risk prediction. *Breast*, **67**, 71–77.
- Rooke,R. *et al.* (2024) The gene ‘degrees of kevin bacon’ (dokb) regulates a social network behaviour in *Drosophila melanogaster*. *Nat Commun*, **15**, 3339.

- Rosenthal,S.B. *et al.* (2023) Mapping the common gene networks that underlie related diseases. *Nat Protoc*, **18**, 1745–1759.
- Shah,S.D. and Braun,R. (2019) GeneSurrounder: network-based identification of disease genes in expression data. *BMC Bioinformatics*, **20**, 229.
- Szklarczyk,D. *et al.* (2025) The STRING database in 2025: protein networks with directionality of regulation. *Nucleic Acids Res*, **53**, D730–D737.
- Vidal,M. *et al.* (2011) Interactome networks and human disease. *Cell*, **144**, 986–998.
- Vogelstein,B. and Kinzler,K.W. (2004) Cancer genes and the pathways they control. *Nat Med*, **10**, 789–799.
- Wang,S.-H. *et al.* (2021) Diagnosis and treatment of an inborn error of bile acid synthesis type 4: A case report. *World J Clin Cases*, **9**, 7923–7929.
- Wu,H. *et al.* (2020) Phenotype-to-genotype approach reveals head-circumference-associated genes in an autism spectrum disorder cohort. *Clin. Genet.*, **97**, 338–346.
- Wu,J. *et al.* (2009) Integrated network analysis platform for protein-protein interactions. *Nat Methods*, **6**, 75–77.
- Yurov,Y.B. *et al.* (2011) The DNA replication stress hypothesis of Alzheimer's disease. *ScientificWorldJournal*, **11**, 2602–2612.
- Yu,W. *et al.* (2008) Gene Prospector: an evidence gateway for evaluating potential susceptibility genes and interacting risk factors for human diseases. *BMC Bioinformatics*, **9**, 528.